Toronto Metropolitan University

Faculty of Engineering & Architectural Science

Course Title:	Computer Programming Fundamentals
Course Number:	CPS 188
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Instructor	Dr. Sadaf Mustafiz

Assignment/Lab Title:	Term Project - Report

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We affirm that this project is original and is our own work

1 Introduction

The purpose of this project is for students to apply their knowledge of the CPS188 - Computer Programming Fundamentals course to a real-world scenario. Particularly, students will familiarize themselves with strings in C and an external plotting software called GNUplot. They will be introduced to new forms of data in which they will write scripts using the different functions available to them in GNUplot. These scripts will then be sent to GNUplot to construct graphs. For this, students will be provided weather information (in comma-separated format) gathered on a monthly basis by different weather-monitoring systems around the world relevant to the 18th, 19th, 20th and 21st centuries. Using their knowledge of C libraries, strings, arrays, loops, and file I/O, they will be required to incorporate the data provided in their C programs to ultimately make calculations to then export and plot graphs based on a variety of proposed questions. The solutions to these questions are briefly outlined in this report. This report intends to share important information regarding weather and climate patterns over the past three-to-four centuries which will be beneficial to individuals studying the Earth's atmospheric data such as meteorologists, geoscientists, and climate advocates. Overall, this report provides students with an opportunity to show their knowledge of the fundamentals of the C-programming language, meanwhile also helping one to to take a global stance on the situation of the Earth's climate.

2 Solutions to Program Requirements

Question 01

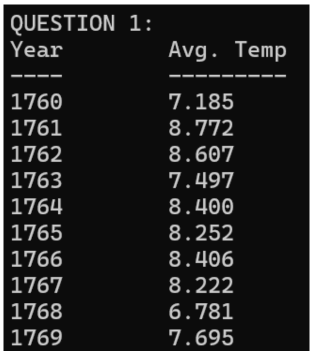
Yearly Avg. Temperatures from 1760-2015

Cut/Paste Terminal Output:

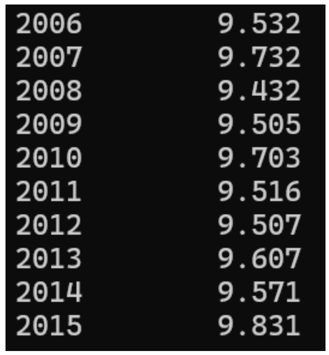
First 10 Years	Average Temperature	Last 10 Years	Average Temperature
1760	7.185	2006	9.532
1761	8.772	2007	9.732
1762	8.607	2008	9.432
1763	7.497	2009	9.505
1764	8.400	2010	9.703
1765	8.252	2011	9.516
1766	8.406	2012	9.507

1767	8.222	2013	9.607
1768	6.781	2014	9.571
1769	7.695	2015	9.831

Code Execution:



```
QUESTION 1:
Year        Avg. Temp
----        ---------
1760        7.185
1761        8.772
1762        8.607
1763        7.497
1764        8.400
1765        8.252
1766        8.406
1767        8.222
1768        6.781
1769        7.695
```



```
2006        9.532
2007        9.732
2008        9.432
2009        9.505
2010        9.703
2011        9.516
2012        9.507
2013        9.607
2014        9.571
2015        9.831
```

Explanation:

Initially, the years 1750-1759 of the csv file were skipped using a while loop condition as requested by the question. Then, starting at the year 1760, the end of file (EOF) function was used with a while loop to iterate through the remaining years. For each year, the average of the twelve months were calculated. Then, the year and the yearly average land temperature was printed to the terminal.

Question 02

Average Land Temperatures for Different Centuries

Cut/Paste Terminal Output:

Century	Avg. Temperature
18	8.215
19	8.009
20	8.638
21	9.542

Code Execution:

```
QUESTION 2:
Century      Avg. Temp
-------      ---------
18           8.215
19           8.009
20           8.638
21           9.542
```

Explanation:

Initially, the years leading up to 1760 were skipped using the fgets() function and a while loop, checking the year. A set of four loops were then used in order to calculate the average land temperature by century. This was done by using while loops leading up to the years 1799, 1899, 1999, and 2015 (using the EOF function), for the 18th, 19th, 20th, and 21st centuries respectively. In each while loop, a running total containing all the monthly averages leading up to the final year of the century, and a counter containing the months accounted for in the century. These were used to calculate the century average (total / months accounted for). These averages were then printed on the terminal.

Question 03

Monthly Average Land Temperatures from 1900-2015

Cut/Paste Terminal Output:

Month	Avg. Temperature
January	2.815
February	3.331
March	5.396
April	8.537
May	11.395
June	13.540
July	14.449
August	13.958
September	12.167

October	9.527
November	6.223
December	3.812

Code Execution:

```
QUESTION 3:
Month        Avg. Temp
--------     ---------
January      2.815
February     3.331
March        5.396
April        8.537
May          11.395
June         13.540
July         14.449
August       13.958
September    12.167
October      9.527
November     6.223
December     3.812
```

Explanation:

Initially the years leading up to 1900 were skipped using the fgets() function and a while loop. Using the EOF function, a while loop running until the year 2015 was created in order to calculate the monthly averages for each month in this timespan. An array of 12 elements was created, one for each month, that would contain a running sum for each month over the timespan. A for loop in the while loop, running for 12 months, added each monthly sum to their corresponding running total in the array. Then, by dividing each sum in the array by 12, the monthly averages were obtained and printed in the terminal.

Question 04

Hottest/Coldest Month Recorded

Cut/Paste Terminal Output:

QUESTION 4:
Hottest Month: 1761-07

Coldest Month: 1768-01

Code Execution:

```
QUESTION 4:
Hottest Month: 1761-07
Coldest Month: 1768-01
```

Explanation:

Initially, the years 1750-1759 of the csv file were skipped using a while loop condition as requested by the question. Two integer variables high and low stored the highest and lowest temperatures recorded respectively. Then, starting at the year 1760, the end of file (EOF) function was used alongside a while loop to iterate through the remaining years. For each year, a for loop was used to iterate through the 12 months. If the temperature of a particular month was greater than that of the “high” variable, it would be updated and vice versa for the “low” variable. Then, the hottest and coldest months along with the respective year were printed to the terminal.

Question 05

Hottest/Coldest Year Recorded

Cut/Paste Terminal Output:

QUESTION 5:
Hottest Year: 2015
Coldest Year: 1768

Code Execution:

```
QUESTION 5:
Hottest Year: 2015
Coldest Year: 1768
```

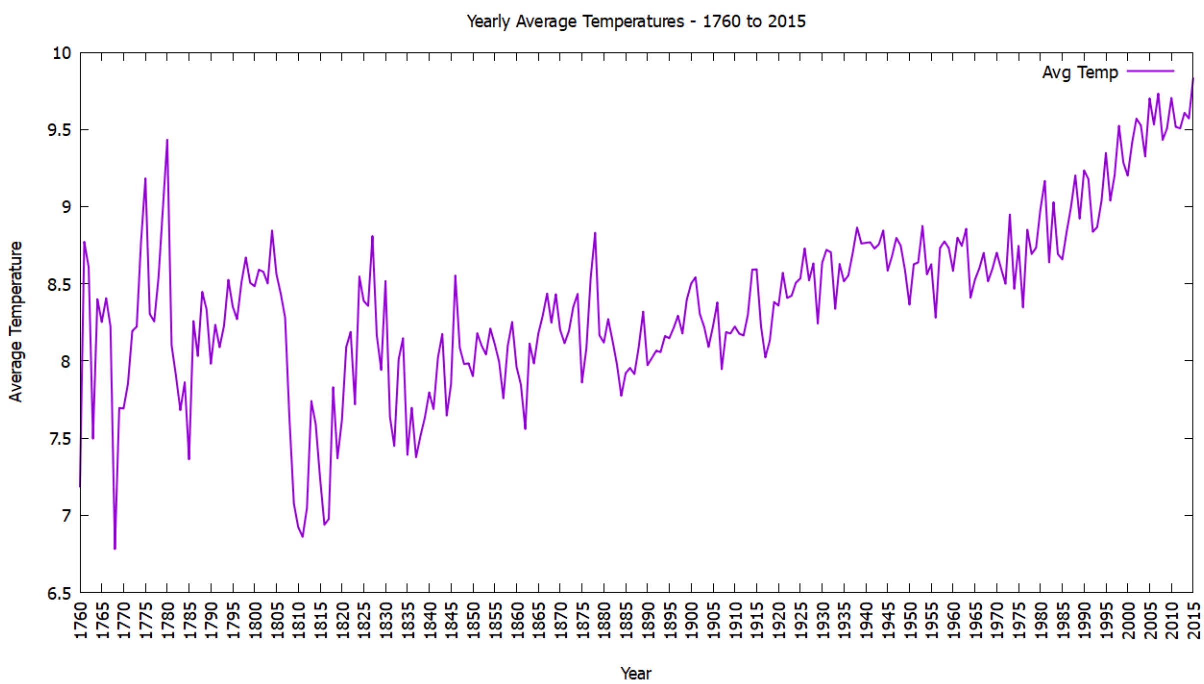
Explanation:

Question 5 was done inside of question 1 because of their similarity. As mentioned previously, the end of file (EOF) function was along a while loop to iterate from the year 1760 to 2015. For each year, a for loop was used to iterate through the 12 months, and the average land temperature of these months were calculated. If a new higher or lower average land temperature was found, the high and low variables, which were used in the preceding question, were updated. Then, the hottest and coldest years were printed to the terminal.

Question 06

GNUPlot - Yearly Temperatures from 1760 to 2015

GNUplot Generated Graph:



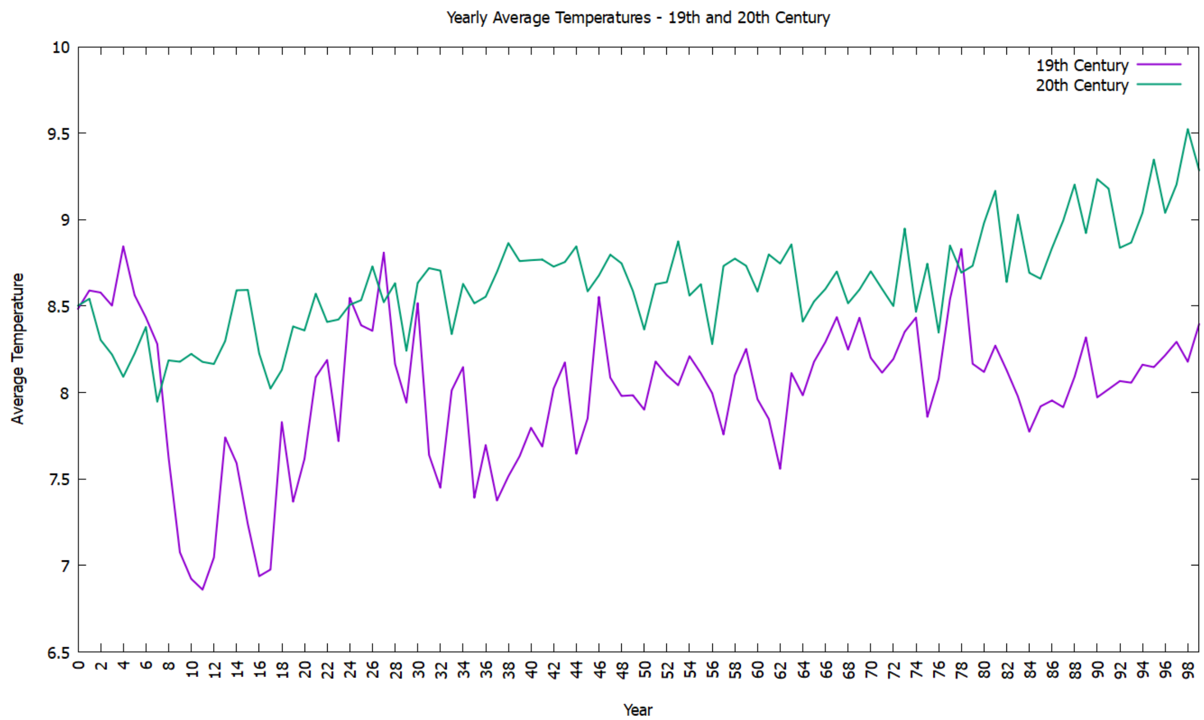
Explanation:

Firstly, the data style was set to be a line plot. Then, the x-y axis was labelled and the graph was given a title. The x-tics were rotated 90° to make it more reader friendly. Additionally, the x-axis was configured such that it ranged from the year 1760 to 2015 and had divisions at every five years. Finally, a legend was also added to the top-right corner of the graph.

Question 07

GNUPlot - Yearly Temperatures for 19th and 20th Century

GNUplot Generated Graph:



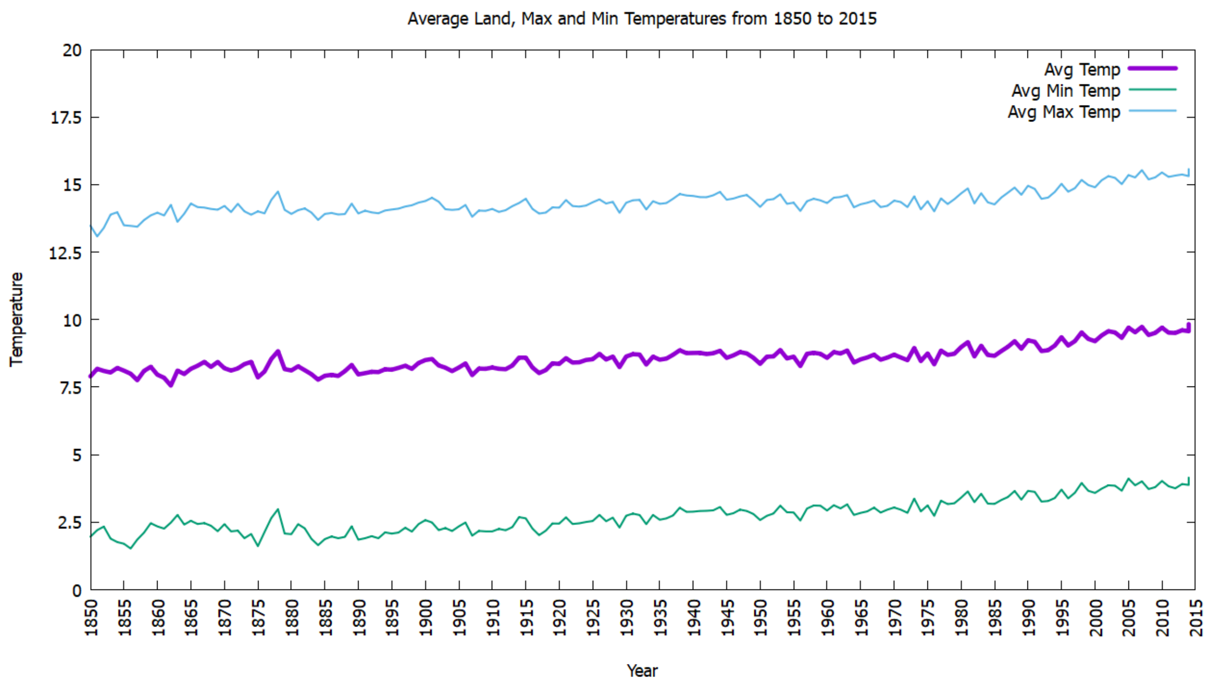
Explanation:

Firstly, the data style was set to be a line plot. Then, the x-y axis was labelled and the graph was given a title. The x-tics were rotated 90° to make it more reader friendly. Additionally, the x-axis was configured such that it ranged from the year 0 to 99 and had divisions at every one year. Finally, a legend was also added to the top-right corner of the graph to differentiate between the temperatures of the 19th and 20th century.

Question 08

GNUPlot - Average Land, Min, and Max Temperatures - 1850 to 2015

GNUplot Generated Graph:



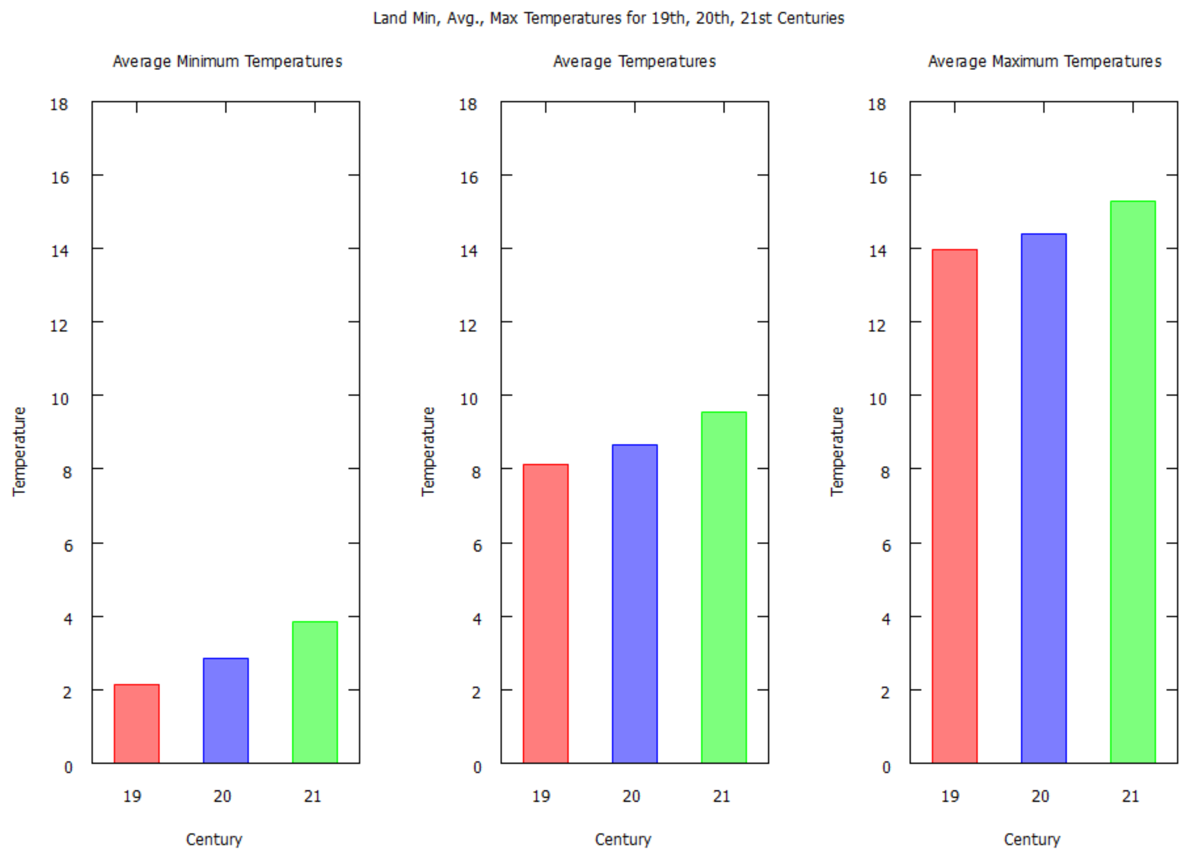
Explanation:

Firstly, the data style was set to be a line plot. Then, the x-y axis was labelled and the graph was given a title. The x-tics were rotated 90° to make it more reader friendly. Additionally, the x-axis was configured such that it ranged from the year 1850 to 2015 and had divisions at every five years. Moreover, the line for the land average temperature data set was made thicker than the other two as requested. Finally, a legend was also added to the top-right corner of the graph to differentiate between the minimum, average, and maximum temperatures for the given time period.

Question 09

GNUPlot - Land Minimum, Average, and Maximum Temperatures - 19th, 20th, 21st century

GNUplot Generated Graph:



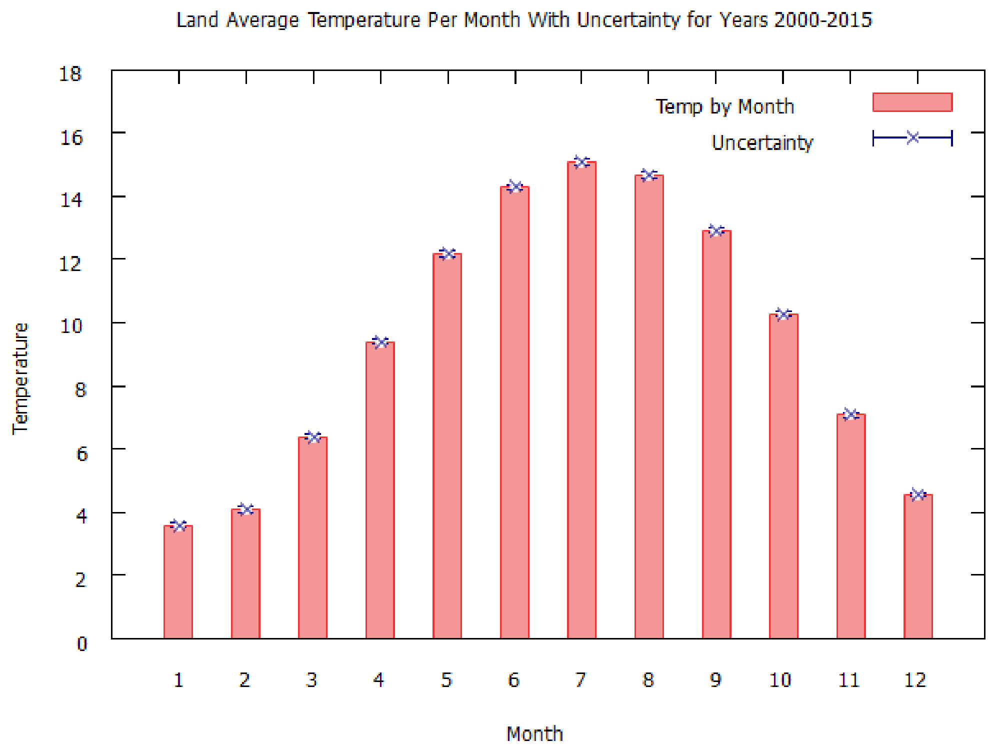
Explanation:

Firstly, the data style was set to be a bar (box) plot. The width and opacity of the boxes were adjusted. Then, the x-y axis was labelled and the graph was given a title. Additionally, the x-axis was configured such that it ranged from the year 0 to 13 and had divisions at every century. Finally, a legend was also added to the top-right corner of the graph to differentiate between the temperature and associated uncertainty.

Question 10

GNUPlot - Average Land Temperature by Month with Uncertainty (2000-2015)

GNUplot Generated Graph:



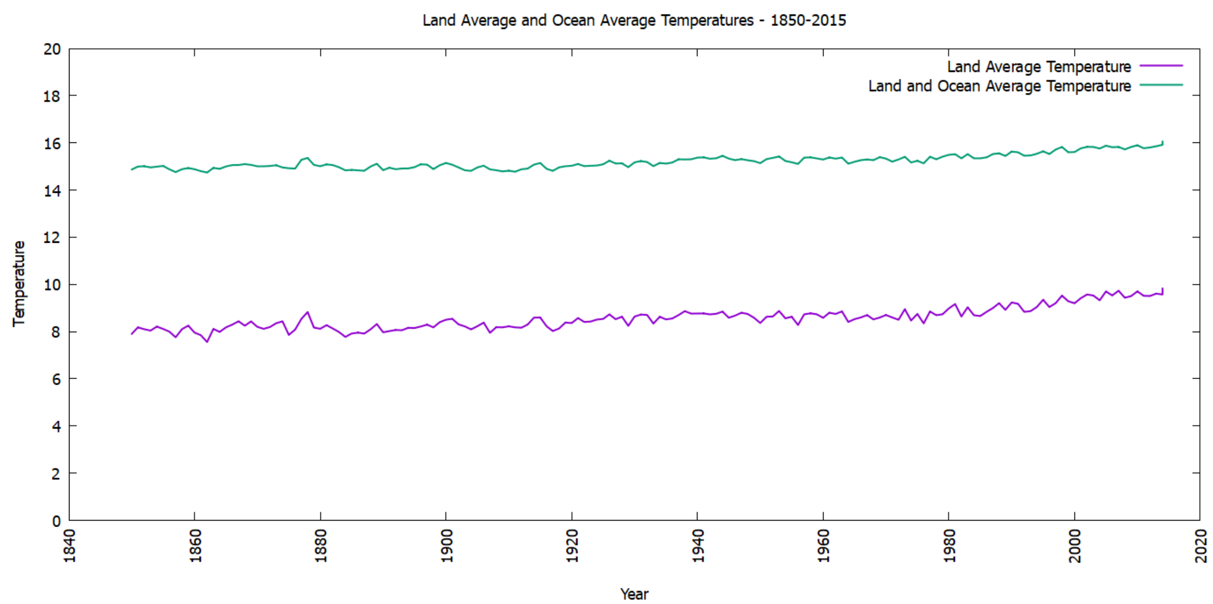
Explanation:

Firstly, the data style was set to be a bar (box) plot. The width and opacity of the boxes were adjusted. Then, the x-y axis was labelled and the graph was given a title. Additionally, the x-axis was configured such that it ranged from the year 0 to 13 and had divisions at every month. Finally, a legend was also added to the top-right corner of the graph to differentiate between the temperature and associated uncertainty.

Question 11

GNUPlot - Average Land and Land+Ocean Average Temperatures from 1850-2015

GNUplot Generated Graph:



Explanation:

Firstly, the data style was set to be a line plot. Then, the x-y axis was labelled and the graph was given a title. Additionally, the x-axis was configured such that it ranged from the year 1840 to 2020 (extra years to make the graph more visible) and had divisions at every twenty years. Finally, a legend was also added to the top-right corner of the graph to differentiate between the land average and land/ocean average temperatures.

3 Conclusion and Thoughts

Looking at the data results and plots created in GNUplot, some scientific insights can be made in relation to global temperature patterns. Overall, it would appear that global temperature, both on land and in the ocean, are on the rise, if nothing else, by a few degrees. This trend can be seen over months, years, and even centuries, and is supported by nearly all data relating to temperature averages over time, as seen in question 1, showing an increase in yearly average temperature. As well, question 2 shows an increase in century average temperature. More importantly, this pattern is clearly demonstrated by the plot present in question 6, showing a steady increase in yearly average temperature over time. Further, question 7 clearly demonstrates the difference in average temperature between the 19th and 20th century. Question 8 shows an increase in overall average temperature by century, a trend seen in the average, minimum, and maximum temperature averages. Finally, question 11 demonstrates a steady increase in average yearly land and ocean temperatures.

Interestingly, as shown in question 4, while the coldest month occurs in 1768, the hottest month occurs earlier in 1760. This is likely due to random variation, and is also why the hottest year being 2015 as opposed to the coldest being in 1768 as seen in question 5 isn't a strong indicator of an increasing temperature trend on its own.

Other, more overt insights, such as the summer months being the hottest, while the winter months being the coldest, are demonstrated quite plainly by questions 3 and 10.

Doing this project provided many valuable experiences, and demonstrated that if done again, many things should have been done differently. Working with C on a project of a relatively larger scale like this demonstrated the importance of organization and commenting, especially between versions. As one person may typically work on a version, commenting is crucial in allowing other group members to understand code changes and additions. Readable code is a must when sharing between group members, and bad personal habits that may work for the individual writing the code must be foregone for the betterment of the group. And version control is extremely necessary. As well, not allotting the work between group members or by a schedule was, while not detrimental, still an avoidable hindrance to the overall productivity of the group. For example, having group members assigned to specific questions from the start, to working on the report, to maintain version control, and other tasks could have easily made the project more efficient.

In conclusion, this project provides many valuable insights on data collection and analysis, particularly through the use of graphing in programs like GNUPlot, on climate patterns, and on the experience of working in a group on a coding project of a relatively large scale.